Final Unseen Deployment-Style Evaluation

One frozen model • one frozen threshold • 50,000 legitimate + 500 phishing URLs

Weighted HYB-MS +10% MS-HardGAN-C (seeds 42/43/44)

FINAL RESULT

Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
98.86%	46.22%	91.80%	61.49%	97.62%	84.38%	1.07%	8.20%

OFFLINE CANDIDATE CONFIRMED

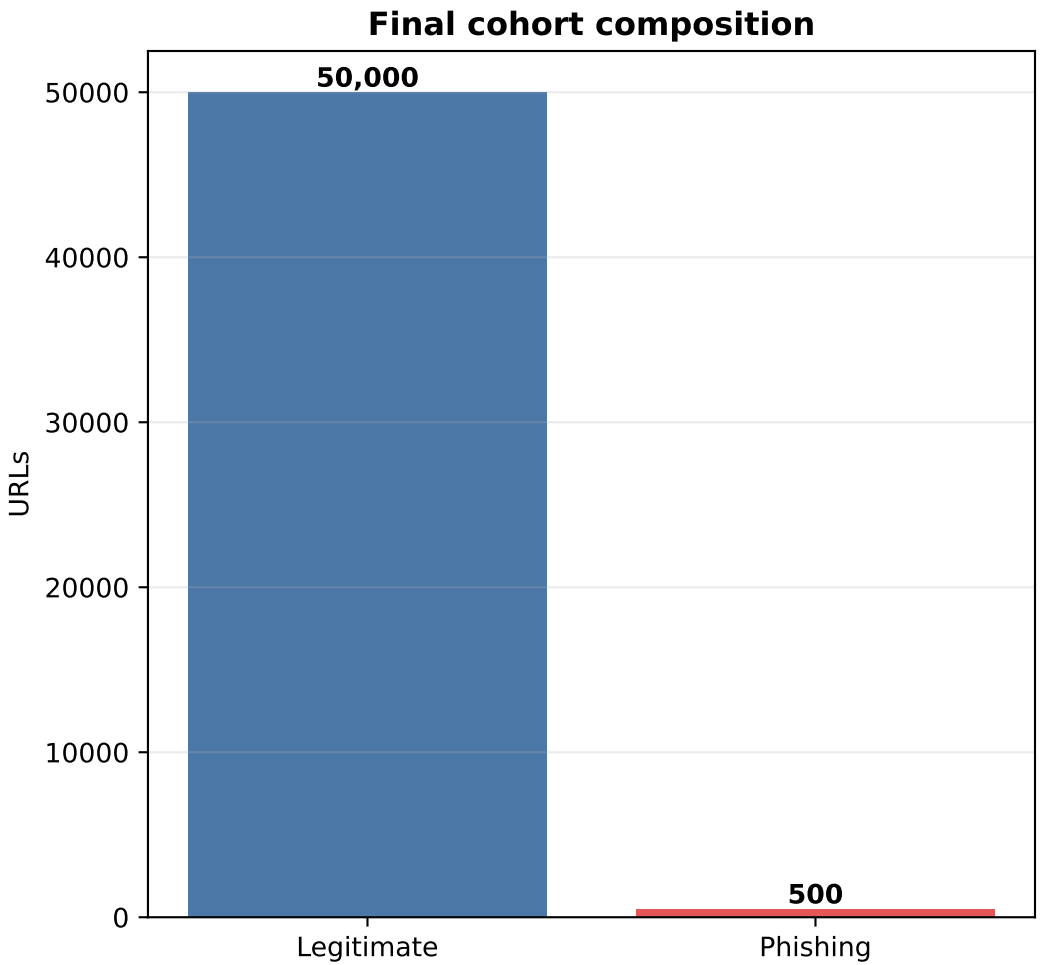
The locked model retained at least 80% phishing recall with no more than 2% false-positive rate on a completely identity-disjoint, low-prevalence cohort.

Confusion counts: TN 49,466 • FP 534 • FN 41 • TP 459

This is confirmation evidence, not permission to tune. The model and threshold must remain unchanged after seeing it.

Why This Resembles Browser Deployment

Low prevalence exposes alert burden that balanced accuracy can hide



Observed prevalence

0.99%

Operational view

False alerts per 10,000 visits: 105.7

Detected phish per 10,000 visits: 90.9

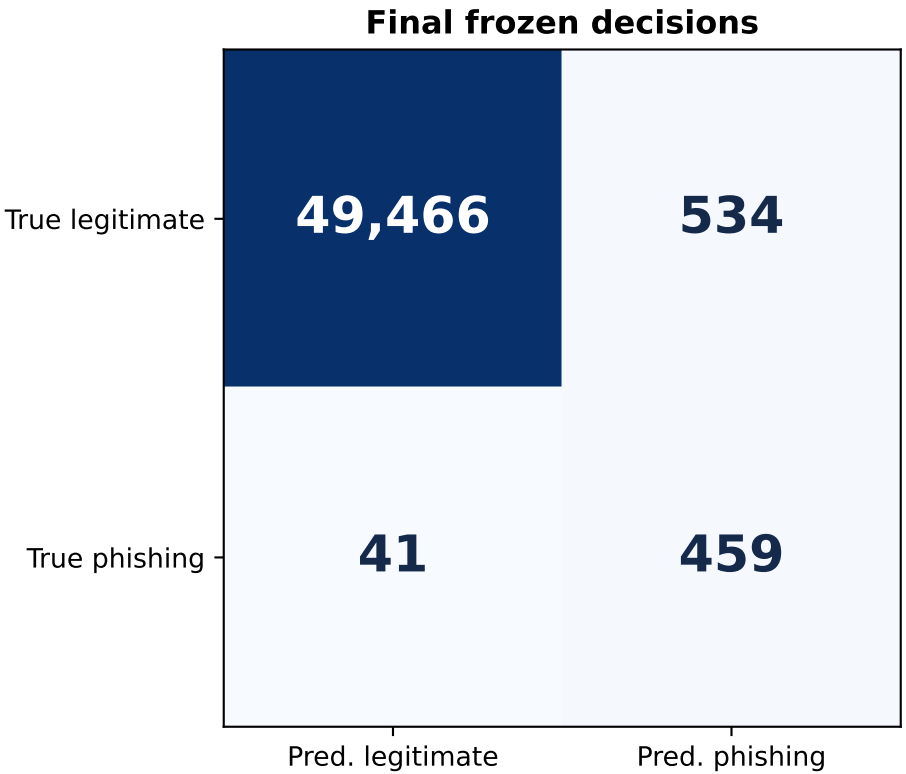
Missed phish per 10,000 visits: 8.1

Alerts that are true phishing: 46.22%

Accuracy alone is misleading here: predicting every URL as legitimate would score 99.01% accuracy while detecting zero phishing. Recall, FP precision, F1, and ranking metrics must be read together.

Confusion Matrix and Uncertainty

Exact counts and 95% Wilson intervals



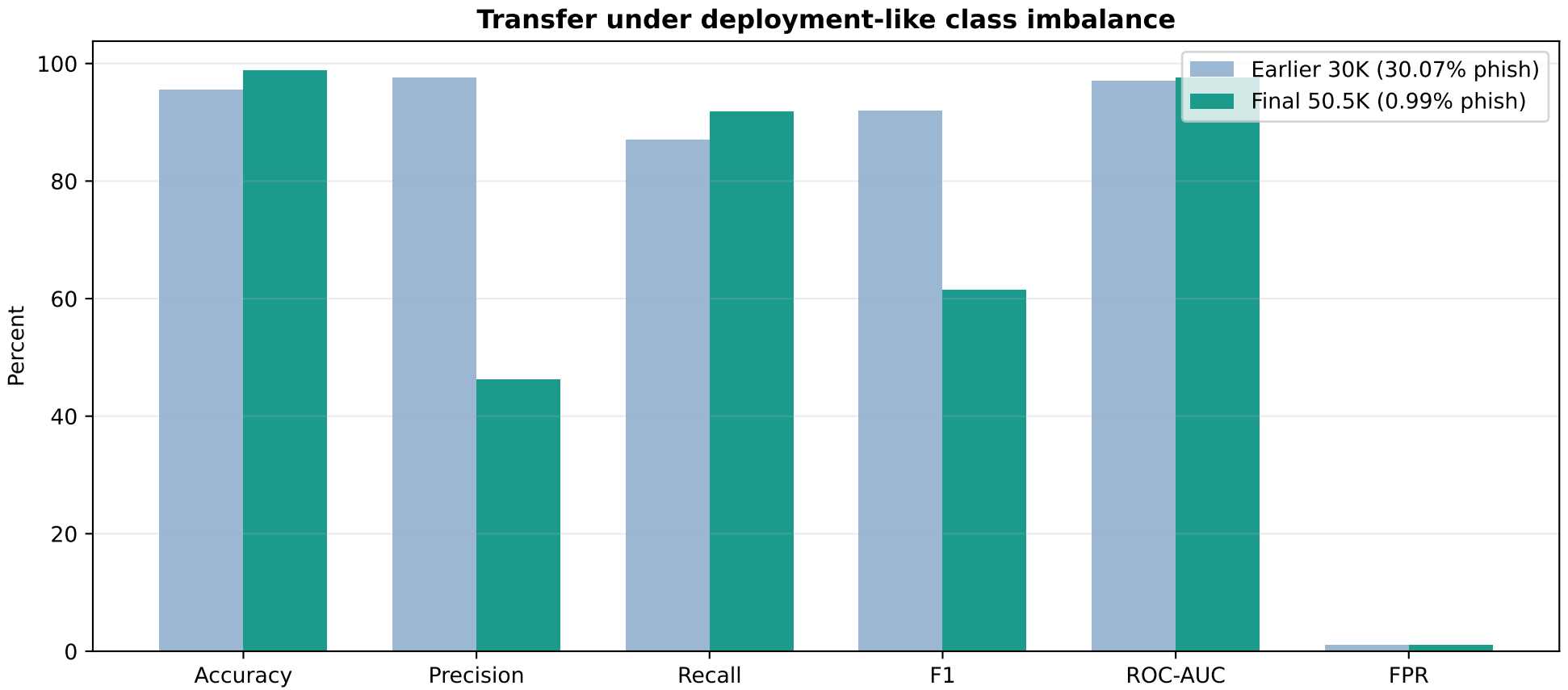
Metric	Estimate	95% CI
Recall	**91.80%**	89.06%–93.90%
FPR	**1.07%**	0.98%–1.16%
Precision	**46.22%**	43.14%–49.33%
Accuracy	**98.86%**	98.77%–98.95%

Specificity: 98.93%
NPV: 99.92%
Balanced accuracy: 95.37%
MCC: 0.6470

The FPR interval is tight because it is estimated from 50,000 legitimate URLs; recall remains less precise because it uses 500 phishing URLs.

Previous Stress Test vs Final Low-Prevalence Test

Same selected model and threshold; different class prevalence and fresh identities



On the final cohort, recall changed from 87.05% to 91.80% and FPR from 0.96% to 1.07%. Precision changed from 97.49% to 46.22%; much of that movement is expected when phishing prevalence falls from 30.07% to 0.99%.

Precision is prevalence-sensitive; ROC-AUC, recall, and FPR are more directly comparable across these two cohorts.

Final Interpretation and Deployment Boundary

What the one-shot confirmation supports

OFFLINE CANDIDATE CONFIRMED

SUPPORTED

- The selected model transfers to a new identity-disjoint cohort at approximately 1% phishing prevalence.
- The result used no retraining, threshold tuning, checkpoint selection, or comparison against the other 105 models.

NOT SUPPORTED

- Production-ready status: latency, memory, privacy, browser integration, monitoring, and live alert adjudication remain untested.
- Universal phishing coverage: two phishing feeds and a Tranco legitimate proxy cannot represent the whole web.

OPERATIONAL NEXT STEP

- Package only this frozen model for a shadow-mode browser pilot; log decisions without blocking users.
- Measure latency and manually review false alerts and missed phish before choosing any production action threshold.